1. ConvertAudioEngine.ps1 *audio-processing script*

Downmixes AAC 7.1, EAC3 7.1, TrueHD 7.1, PCM/FLAC 7.1 to DDP 5.1 (1024k)

- Re-encodes **5.1 audio** to **DDP 5.1 (768k)**
- Encodes **stereo audio** to **DDP 2.0 (256k)**
- Passthrough for **EAC3 Atmos**, **EAC3 5.1**, **EAC3 2.0**, and **TrueHD 5.1**
- Maintains perfect sync on long-duration files
- Normalizes malformed channel layouts
- Removes 2-channel commentary tracks
- Produces clean, predictable, device-safe output

Requirements

- **PowerShell** 5.1 or 7.x (fully tested on both)
- **FFmpeg 8.1 or newer** — Must be in your PATH or the same folder as the script. You can grab the latest builds from the official FFmpeg download page.

1. Place `ConvertAudioEngine.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

```
powershell -ExecutionPolicy Bypass -File .\ConvertAudioEngine.ps1 ".\YourMovie.mkv"
```

🎧 Supported Audio Codecs

The engine supports and processes the following codec families:

Codec Family	Formats Included
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm*s16le, pcm*s24le, pcm*f32le, pcm*f32be
FLAC	All FLAC channel layouts

---

2. FFmpeg Global Options:

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- `-threads 8` * User-adjustable (4–16), 0 = automatic
  - **What:** Sets the number of CPU threads the encoder is allowed to use.
    - **Why:** Helps balance encoding speed and system stability. It prevents CPU saturation that can lead to frame drops or system instability during the muxing process.
- `-analyzeduration 200M & -probesize 200M`
  - **What:** Increases the initial data analysis buffer to 200 Megabytes.
    - **Why:** High-bitrate audio like TrueHD with Atmos metadata or AAC 7.1 requires a deep probe to fully parse the channel layout. Standard 5MB,10MB,20MB probes often misidentify these tracks, causing the encode to fail or use the wrong channel map.
- `-err_detect ignore_err`
  - **What:** Instructs FFmpeg to bypass minor bitstream errors.
    - **Why:** Prevents a single corrupted audio frame (common in digital rips) from aborting a multi-hour conversion process.
- `-drc_scale 0`
  - **What:** Disables Dynamic Range Compression (DRC).
    - **Why:** Ensures the output audio maintains the full dynamic range of the source. It prevents the encoder from "flattening" the sound or lowering the volume of loud peaks.

3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero`
  - **What:** Forces all stream timestamps to start at absolute zero.
    - **Why:** Many digital files contain negative initial "Presentation Time Stamps" (PTS); this clamps them to zero to prevent audio lag or lead.
- `-max_muxing_queue_size 9999` *(1024-100000) recommended range
  - **What:** Increases the muxing buffer headroom.
    - **Why:** Pass+Copy mode creates 3 streams (video copy, 7.1 copy, 7.1→EAC3 encode). Two copy streams flood the muxer while the encode pipeline catches up; 9999 buffer prevents de-sync on 3+ hour files.
- `-map 0:v? -c:v copy`
  - **What:** Maps all video tracks and sets them to stream-copy mode.
    - **Why:** Ensures zero quality loss for the video while moving at maximum speed.
- `-map_metadata 0 -map_chapters 0`
  - **What:** Copies global metadata and chapter markers from the source.
    - **Why:** Preserves the original file's organizational structure (Title, Year, Scene markers).

4. Stereo (2.0) Optimization: Dolby Pro Logic II Compatibility

For stereo tracks, the engine uses specific flags to ensure they playback correctly on both standard speakers and surround sound systems.

- **-dsur_mode 1**
  - **What:** Enables the Dolby Surround Mode flag in the EAC3 bitstream.
    - **Why:** Tells the playback receiver that the 2.0 track contains matrixed surround information (like Pro Logic II), allowing it to properly expand the soundstage.
- **-stereo_rematrixing 1**
  - **What:** Explicitly enables rematrixing during the stereo encode.
    - **Why:** Ensures that the left and right channels are balanced correctly during the conversion from higher-channel sources to stereo.

---

5. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- `-filter:a:$i $panFilter`
  - **What:** Applies the custom channel-mapping matrix (the Pan Filter).
    - **Why:** Necessary for downmixing 7.1 or 5.1 to smaller layouts. It ensures that "lost" channels (like back-surrounds) are projected into the remaining speakers rather than simply deleted.
- `-c:a:$i eac3`
  - **What:** Sets the codec for audio stream `$i` to Dolby Digital Plus (EAC3).
    - **Why:** Chosen for its high efficiency and broad compatibility with modern smart TVs and home theater receivers.
- `-b:a:$i $t.Bitrate`
  - **What:** Sets the target bitrate (e.g., 1024k for 5.1, 256k for 2.0).
    - **Why:** High bitrates ensure the "transparency" of the encode, making it difficult to distinguish from the lossless original.
- `-dialnorm -31`
  - **What:** Sets Dialogue Normalization to the reference "off" position.
    - **Why:** Prevents the playback device from applying its own internal volume leveling, preserving the original mix's intended loudness.

- `-cutoff 20000`
  - **What:** Sets the low-pass filter frequency to 20kHz.
    - **Why:** Preserves the high-frequency "clarity" of the audio that standard encoders sometimes discard to save space.
    - **Applies to:** Downmix, 5.1 re-encode. Not applied to the stereo (2.0) encode path.
- `-metadata:s:a:$i "title=..."`
  - **What:** Injects a custom string into the stream title metadata.
    - **Why:** Allows users to see exactly what the track is (e.g., "DD+ 5.1 Downmix") in their media player's audio selection menu.

---

6. Audio Rules & Priority Logic

The engine uses a **Priority Mapping (0-100)** to evaluate tracks based on quality and codec type.

7. Audio Rules & Priority Logic

EAC3 / Atmos Rules (Priority 100)

Condition	Action	Output	Notes
Atmos (JOC)	Passthrough	Original EAC3 Atmos	Preserves object metadata.
EAC3 7.1	**Downmix**	DDP 5.1 @ 1024k	Pri 99: Maps 8 channels to 6 using Pan Filter.
EAC3 5.1	Passthrough	Original EAC3 5.1	Pri 98
EAC3 2.0	Passthrough	Original EAC3 2.0	Pri 97

TrueHD Rules

Channels	Action	Output	Priority
5.1	Passthrough	Original	81
7.1	Downmix	DDP 5.1 @ 1024k	80

AAC Rules

Channels	Action	Output	Priority
7.1	Downmix	DDP 5.1 @ 1024k	92
5.1	Encode	DDP 5.1 @ 768k	91
2.0	Encode	DDP 2.0 @ 256k	90

DTS Rules

Variant	Action	Output	Priority
DTS-HD MA/HRA (multichannel)	Encode	DDP 5.1 @ 1024k	73
DTS Core (multichannel)	Encode	DDP 5.1 @ 768k	72
DTS-HD 2.0	Encode	DDP 2.0 @ 384k	71
DTS 2.0	Encode	DDP 2.0 @ 256k	70

PCM / FLAC Rules

Channels	Action	Output	Priority
7.1	Downmix	DDP 5.1 @ 1024k	62
5.1	Encode	DDP 5.1 @ 768k	61
2.0	Encode	DDP 2.0 @ 384k	60

---

8. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

The Formula

`aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707*SL|BR=BR+0.707*SR`

Layout Pinning (aformat)

What: Forces the decoder to output a strict 8-channel 7.1 layout *before* the math is applied.
Why: Atmos metadata or variable codecs can cause "amorphous" decoding where channel positions shift. **Pinning** ensures the Pan filter's "Side Left" label actually points to the side surround data.

The Math (0.707)

What: Applies a **-3dB** coefficient to the Side Surrounds (SL/SR).
Why: Prevents digital clipping and ensures the side-channel sounds don't overpower the rear soundstage.

---

9. Commentary Filtering Logic

Only 2-channel commentary tracksare removed to maximize space without risking primary content.

- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** `commentary, director, producer, writer, cast, behind, bonus, alt, interview`.
- **Why 5.1 Commentary is NOT removed:**
  - It is extremely rare in retail media.
    - It is often hard to tell apart from legitimate "Alternate Mixes" or "Home Theater Optimized" tracks.
    - Removing it carries a high risk of "False Positives" (deleting real movie content).

---

10. Malformed Layout Guards

The script corrects illegal 7-channel layouts:

- **AAC 7ch** → 6ch, **EAC3 7ch** → 6ch, **TrueHD 7ch** → 8ch, **PCM/FLAC 7ch** → 8ch
- **Dialnorm (-31):** Sets Dialogue Normalization to -31dB. This is the "off" setting for Dolby metadata, preventing the playback device from applying its own "loudness leveling" and ensuring you hear the audio exactly as encoded.

⛔ Fallback Rules

If no specific rule matches, the following logic is applied:

Channels	Action	Bitrate
≤2ch	Encode	256k
3–6ch	Encode	768k
>6ch	Downmix	1024k

---

11. Summary Engine

At = Audio processing Summary =, the script displays a table to audit the engine's work., the script displays a table to audit the engine's work.

Column	Description
Index	The original stream number (1, 2, 3,...).
Codec	The source format (e.g., TrueHD, DTS).
Channels	The detected channel count (e.g., 7.1, 5.1).
Action	**Green**: Passthrough, **Yellow**: Downmix, **Blue**: Encode, **Red**: Removed.
Output	The final codec and bitrate configuration.
Priority	The rank assigned by the engine (0-110).
Rule	The specific logic match (e.g., "TrueHD 7.1 Rule").